

"Falcon 2.0T: Cylinder misfire under load from cracked ignition coil"

Document: tsb-20-501

Area: bulletins

1 Condition

Owners of affected Falcon 2.0T vehicles may report a rough idle, hesitation, or stumble under load accompanied by an illuminated check engine light. Scan tool retrieval typically shows random misfire code P0300, often together with cylinder 1 misfire code P0301. The condition is usually more pronounced under high load or in damp conditions. General misfire diagnosis is covered in Falcon Engine Service Manual :: Misfire Diagnosis .

2 Cause

The root cause is a hairline crack in the ignition coil housing of the original-fitment coil IC-2042-A, which allows spark flashover and intermittent misfire. The resulting energy loss sets P0300 and the cylinder-specific P0301 when the affected coil is on cylinder 1. Moisture intrusion through the crack aggravates the misfire, explaining the wet-weather correlation. Coil identification and the related supersession are detailed in Heron Ignition Service Manual :: Coil Identification and Supersession .

3 Repair Procedure

Begin with misfire diagnosis procedure PROC-MISFIRE-DIAG to confirm the misfiring cylinder and rule out fuel and mechanical causes; the full diagnostic flow is in Falcon Engine Service Manual :: Misfire Diagnosis and the misfire code reference is in Harrier Engine Service Manual :: Misfire Diagnostic Trouble Code Reference . Replace the cracked coil IC-2042-A with the superseding part IC-2042-B following coil replacement procedure PROC-IGN-COIL as documented in Heron Ignition Service Manual :: Ignition Coil Replacement Procedure . Inspect and, if fouled, replace spark plug SP-1108 to the published specification per Heron Ignition Service Manual :: Spark Plug Selection and Specifications . Clear P0300 and P0301, then road test under load to confirm the misfire is resolved; broader ignition system reference is available in the Heron Ignition Service Manual .